

ActInf Livestream #029.2 ~ "Active Inferants: An Active Inference Framework"

Livestream | 1:41:20

welcome to the active inference lab live
stream number

29.2

today is september 28

2021 and we will be discussing the paper

active in france

an active inference framework for ant
colony behavior and today we are here

with the first author daniel friedman

the active inference lab is the

participatory online lab that is

communicating learning and practicing

applied active inference you can find us

at all of the links shown here

this is a recorded and an archive live

stream so please provide us with

feedback so that we can

improve upon our work

all backgrounds and perspectives are

welcome here and we will be following

good video etiquette for live streams

here at the short link you can find all

of the upcoming and recent live streams

that we have done so please get in touch

if you would like to participate

today our goal is to discuss the paper

um that we just said active in france

and we are going to um

do some different papers in the upcoming

weeks like i said get in touch if you

want to participate

so

we're just going to introduce ourselves

and go through a little bit of warming

up i am blue knight i'm an independent

research consultant from new mexico and

i'll pass it to

thanks blue for facilitating i'm daniel

and i'm a researcher in california

and i guess i'll

start with what i'm excited about today

i was just excited that in 2091 we could

sort of forage out in a lot of different

directions and make some initial trails

to a few different ideas

and then now in the 29.2

maybe we can develop some of those

threads splash trails and maybe uh get

some of it down in writing so it should

look um

pretty good

nice yeah i'm excited too um there are

some questions that i've been rolling

over over the last week

and so i'm kind of excited to

delve into those and also into the code

for the model today if we have time um

being that it's just us i think we might

get there well maybe though maybe more

people will join

there might be a little bit of lag

i'm not sure

so you might want to check that

okay

yeah

um

i don't know if it's my internet

um

so the big question of this paper is um

how can a group of active inference

agents

solve complex group foraging problems

well first what i'll say is if you're

recording it locally as well as
streaming then
we can always upload a high quality
recording
and it does sort of bear on this big
question this is a quote from the paper
and that's sort of the big question in
collective behavioral studies which is
okay if greeting stephen if each agent
if each subunit of the system had all
the information needed well then there
wouldn't necessarily be a challenge for
coordination it's like if it already
knew what to do then
it would be a challenge but how do
systems make adaptive decisions when
subunits don't have all the information
so in the case of an online team each
person's only seeing a different
part of the picture in the case the ants
they're each in a little bit different
part of their niche but those are some
of the similarities which is that each
nest mate is only going to be
experiencing its local environment
and
we want to have colonies that work
so welcome stephen maybe you can say
hello while blue's just testing some
stuff out
oh great yeah good morning um
well in north america anyway um yeah i'm
based in toronto um
and i
work with communities and
sort of systemic coaching approaches and
very interested in how active influence
scales in different ways so

even though this is about small things
it's also about lots of small things
so
well your audio also just had some
weirdness steve maybe talk directly into
it
oh okay is that better yeah
oh okay but you mentioned the scaling
and small the one one funny thought
again this picture is from the ants
that i was studying for five years in
arizona
and
um yes the nest mate looks
like a tiny guy like a small ant
but then
we know that these colonies can live for
20 30 plus years
they're also several meters
under the ground
and they weigh you know dozens of
kilograms so then i'm thinking okay it's
a similar age
weight and height to myself
and it's sub-use but
let's see how's that is it any better
it still is not doing anything i just
reopened it but if it's not yeah then i
guess okay now it's uh just the slides
or just a jitsi so it looks like it's
working now okay perfect yeah it's the
slides open also it just is too much um
yeah
yeah welcome back everybody fun times
the show goes on
foraging never stops so it does and
steven will rejoin us hopefully he's
having audio issues also it's a good day

for technical uh mishaps we're having
rainy weather in new mexico which is
never ever happens so all the ants are
underground hiding and the internet does
not want to cooperate

apparently this is so

all right so speaking of multi-scale uh
integration and and across systems as
happens in ants and in the ant colony um
so we we talked about making this table
last time

and so here like what how is each system
related to

the agent the niche the interaction and
stigma v

so we started this this table do you
want to uh maybe take us through what
you've started here daniel

yeah um

see if you can

[Music]

get the slides again because it's not
going to be visible at all on the stream
oh okay i can also switch yeah i can
switch the also yeah

um so

a lot of informal analogies and
connections were being drawn amongst
different systems

and that's kind of what complexity and
systems thinking and active inference is
about like finding patterns across
systems so

maybe there's other ways to break it up
but at least one way we could at least
start to classify

and find similarities and differences
between different systems would be

this way with the columns so we have the
system of interest like the ant colony
as a system of interest
then we can think about what is the
agent level what is the sub unit level
and
that might be the level where the active
inference agent is being specified
we then have the niche that the agent is
operating in so that's the generative
process
that's presenting observations or
outcomes to
the agent which is embodying a
generative model so that's related to
this generative process versus
generative model distinction
the process is the
the process actually outputs
the like temperature or the observations
whereas the generative model is a model
of the process
and then there's two kinds of
interactions that are possible
so there's sort of a transient
interaction
that might be amongst agents welcome
back stephen there's the transient
interactions amongst agents
and in the case of the ants that's like
an antennal contact and then there's the
stigmatic
interactions
where there's a modification of the
niche
and so
sometimes it helps to think about these
interactions amongst agents that

are um without memory as being like the
transient horizontal interactions like
the common example of the bird flock
those are the birds looking at each
other their interactions amongst agents
that modify the likelihood of agents
doing certain future behaviors but they
don't leave a trace on the environment
and that can be contrasted with
stigmatic
or niche modification behaviors which
are defined by being the ones that do
leave a trace on the environment so
maybe we'll
explore different sections today
but we can always say okay let's return
to slide eight and let's add a row
what's a system of interest let's add a
column or let's fill in some columns so
so far we have ant colony brain and
internet slash computer abcs
get the systems that matter um steven
what what would be um a fun system to
throw into this mix and to look for
parallelisms among
we can't hear you steven
yeah we still don't hear you steven
yeah your microphone's definitely not
working
okay
so i've started filling in for for brain
and you know it all really is so
dependent on what the
what level you want to evaluate right so
you know on one level there's like the
little sub regions of the brain and you
took us through i think in the dog zero
video a very nice um

example for ants that has been kind of
done describing the different like
organoids or substructures within the
brain because definitely there are
structures that all have
different functions like the cerebellum
is motor function primarily and you know
the cortex is like higher level thought
and and so there's you know these
different sub regions of the brain but
then also at the cellular level there
are interactions you know that that
expand those regions that are from you
know the cellular cellular level um
and what is the so what the niches or
the interaction
um you know might might be different
depending on like the the cells are
responsible for driving the interactions
between the organoids
so i'm not sure what the level of um
of
the level of analysis the right one is
and then stigmatizing modification i
think about you know long-term
potentiation
um and like adaptation within the brain
is very um
maybe not stigmatic but it's a it's a
really important modification i think
um and sensitivity to different
neurotransmitters and
things like that
blue it's still really lagging can you
just stop streaming and i'm gonna pick
it up on my side okay perfect go for it
so stop the streaming and then just let
me know when you did it let's see how

this

okay let's see what happens

this could be very very interesting

to do the live hack like this

okay youtube we're gonna wait for it to

catch up sorry everybody um

so it goes okay

wow we just hot swapped

live stream feeds

so we're back in the game

no level of preparation that can prepare

for laggy internet i suppose no and it's

like half the time of mine and a half a

time on your side but um

if you're watching live definitely add

any comments or questions because we're

just going to be kind of exploring and

chilling today

but continue to um facilitate and i'll

just be broadcaster role

cool so the um

[Music]

yeah we were talking about the brain and

the interaction and the sigmoidy and

then for the internet

i don't know what do you think is the

is the agent or what what is the sub

unit of the internet like is it each

server or

where do you have the sub unit at

daniel or stephen what do you guys think

well so this this sort of ties into what

they might say in in research you know

unit of analysis as well but that's not

necessarily quite the same

thing as an agent um

i i suppose on the internet

ultimately

it's the um
it's it's whoever's um
making
who's that whoever's triggering choices
um
in the system
um which is normally a yeah it's
normally a person
um
interacting with the screen in some way
i think the overwhelming amount of
interactions on the internet are
probably not human driven
because any human
interaction
is going to
trigger truly a vast number of
communications
amongst other systems
inside of the computer and probably
remotely
so maybe
and i think this is one of the unique
uh challenges but also interesting areas
is like
the internet is a cyber physical system
so we could have one model where the
agent like the subunit are physical
devices so we kind of draw
the blanket
we um
focus on the partitioning
around a physical object like internet
of things we have partitionings around
things
we could also have a human-centered
partitioning so partitioning around
humans but then there's a lot of

gray space in between the humans
a lot of links that aren't human between
the human links
and then another approach might be
purely informational or computational
and seeing the agents as software agents
but i think that
interface where how will we talk about
the
software and the hardware agents coming
together is where
the internet of things is
developing and what it's coming to mean
i mean one thing you could think about
there also is when say for instance i
make a choice or i do something
on the internet
and there's a whole load of there's a
cascade of other things going on through
the different routers and the servers
etc is that more like the pheromone
trail
that sort of dissipates off into the
distance leaving records
and how much of its like other agents
initiating because often
you know
every time you do something to some
extent yeah databases might update
um
however
and this may change
to what extent
that is alive is another question
you know is it is it sort of a
computational form of niche modification
that's going on in most cases in terms
of the internet

but what they're trying to do because
it's that idea of the battle for your
mind they're trying to make it so that
in most cases that
the user
has less
has less thinking to do
in terms of the choices they make
and the design there's this there's a
great book in web design called don't
make me think
and the idea is that you go somewhere
it's almost obvious what you need to do
a bit like when you pick up a newspaper
you know that you're just going to turn
the pages
so
how
how much of all this
architecture in the background is to
support giving you what you expect to be
given when you expect to be given it
um and how much of it is um
which you know how much of it is
actually um what you might term an agent
with with some sort of
teleology
well
you mentioned expectation there and um
expectation is something that agents
engage in so maybe it's even we could
keep on forking this off but
what do we know from the active
inference ontology what defines an agent
external states blanket states internal
generative model what does the internal
generative model entail well it includes
predictions about external states it

includes policy affordances it includes preferences and includes expectations over some of these things as well so when we say agents we can be referring to the whole markov decision process or other ways of viewing the active agents and then the expectations of the agent are really important because it's not just that psychologically it's going to determine whether the agent likes it or not but it helps us connect these systems like the ant colony and subunits of the brain so we don't need to use only the psychological interpretation but then we can have a psychological interpretation where one seems warranted given the system of interest but also we're going to be able to compare that and look for similar functional features in systems without getting into just some sort of abstract debate over whether it's alive or whether it's certain feels a certain way to be that system so daniel you brought up the generative model and i was just going to bring it up um right before this and so i've been really uh you know thinking a lot about the generative model and like where is the generative model which is like i think probably my favorite podcast episode that i've ever cut is going to come out next week and it talks about like the location of

the generative model and so
you know it doesn't necessarily have to
have a location
because
it's like software right like so i mean
that yeah there's like a copy of the
software running on disk
maybe maybe it's in the cloud like it
does but it doesn't matter because it's
the program that's running so the
location of the program is almost not
even relevant
um so i just i think about like what is
the generative model
of the internet right like what does
that look like
great question um inactivism for cyber
physical systems
and embodiment
uh in encodement maybe there's some
analogous term
to what would describe
what we call 4e in the embodied extended
and cultured
etc
for digital systems
because they do have
important attributes
that
in a strange way almost suggest that
some of these attributes some of these
features that are suggested to be
quintessentially related to having a
body and embodiment
also exist for software
could i putting something on to that
is say we think about rather than and
you've got the active inference

generative model

but say you've got an active niche

modification

algorithm

so ie

it's still

working

off the agents being making choices but

what's happening is so for instance say

you've got abc testing happening on a

website so basically they send out three

different campaigns with three different

types of

profiles

now the user makes choices

based on how the user

makes a choice

they may be

across the whole of the

a whole market

the abc choices might get modified in

that niche but eventually maybe then

actual

clients get sent down certain routes

um i mean a classic case of this is also

what happens with those in-store cards

because i actually i ended up going i

didn't get go for it again i had an

interview with the nectar card people in

the uk once and they uh

they do these cards that you get in

store and basically i think it worked

out they have like a million

permutations of offer that they can give

you based on what your

different buy-in options are so in a way

they're kind of

that but they only serve you up then

whatever you see one of those right you
don't see what all the other people are
getting and some of them are small
changes some of them might be bigger
changes so in some ways it's like it's
still the it's still like it's like the
niche is modifying
to the agent's adaptive behavior
but the actual niche isn't necessarily
doing adaptive active inference but
there may be some sort of active
inference in a more kind of
i don't know non-living way
um

there might be um we'll probably have
better words and ontologically informed
ways of talking about it but there's
kind of like low bar active inference or
like austere active inference that is
just any input output structure
it's going to be able to be fit into
something like
a partially observable markov decision
process so whatever it is that the niche
is doing it's emitting
observations so we can model it under
this certain low bar active inference
versus the uh sort of
anti-dissipative
truly cognitive
strategic active inference
and um

also there's no reason why we can't have
hybrid models where there's like active
inference agents
in a niche that isn't an active
inference agent so like in
this ant model

the niche is just
a landscape of pheromone densities with
a random dissipation factor
so
that
external niche
could be made very
um much more complicated
and the observations that it passes back
to the agents could be like customized
to them
and that could be
calculated by an active inference niche
or not and that's sort of the extremely
interesting
work of axel constant and others
with
some type of symmetry occurring with the
agent and the niche jointly inferring
each other so like what is the niche's
generative model of the organism what is
the internet's generative model
of the user so it's
interesting questions
i love that idea of
symmetry especially between the agent
and the niche um and its relationship or
potential relationship to the kind of
gaia theory
that the the niche has to
expect or have an expectation of the
organism that's going to
interact with it or create it or modify
it or
exploit it in some way um in order for
there to be some kind of reciprocal
um interaction there
so uh super nice

um maybe

we can come back to looking at this um
table and maybe think of other systems
that might be

interesting to go through

and move on to

some of the um

questions

for

29.2 but wait before we go to this one i

want to go to this one hold on

ha

so we were talking about um

the role of dopamine last time which

slide

uh so sorry this is slide

um i skip to

i'm sorry i can't see

um

i think it's number seven

yeah number number eleven

oh wait okay

so um

so we were talking about last time the

role of dopamine

and like influencing the affordances and

and this really kind of speaks to the

role of reward

um

in a generative model and and so you

know we talk about one of the beautiful

things about active inference is it kind

of

changes the reward um structure

in the model

because you know there's some kind of

reward that's given

to

you know actually finding the food right
so so there's some some reward for
finding the food but also and and even
in this model so
the the reward structures has changed
right so there's um it's not just about
finding the food um and maybe this is a
good opportunity to to look through the
code maybe
as well but but you know we think about
the change in the reward structure and
we place some value on the epistemic um
you know the knowledge about the model
of the world so if we can build a more
precise model of the world
we can also um
gain more food overall
so i don't know if uh you guys have
comments on that or yes steven guide
yeah i think also as well as built like
you say building that model of the world
becomes a it's an action model okay so
it can be
what might be seen factually
spurious but practically useful
you know so it may be really useful to
have some bizarre
ritual about how to get on a soccer
pitch and try and kick the ball right
but if it
does something better than
what you're doing otherwise
go for it
and i think the challenge comes with um
increasingly esoteric rituals if they're
not related to the actual causal
structure of the world
if they work for a moment they're not

going to be adaptive because if your only strategy is you know praying to the kicking god instead of practicing it might work or it might indirectly induce some sort of outcome but is that a long-term strategy um but agree that the generative model is including a model of action it's not just about getting precision about how things are out there because what ultimately matters is getting precision on our actions and then there's sort of these two parallel paths or two intertwining paths which is do we take a reward centered approach is behavior dominated by reward and then the other side of the reward coin is like the um punishment aversion so carrot in the stick that's sort of a sister road two-lane road do we pursue reward or do we avoid regret and suffering and then a third road is this uncertainty minimization approach so i pasted in this paper by columbo explanatory pluralism and um they review three prominent hypotheses of dopamine activity so anhedonia which is unpleasurable states incentive salience and reward prediction errors and i i think this is a really fascinating paper and it does apply to the ant context because they make a very

specific

system point

talking about the role of dopamine in
the mammalian brain reviewing empirical
evidence and then also

pull back to the broader implications
for the philosophy of science

so

they conclude that the evidence
currently vindicates explanatory
pluralism in other words these
explanations do not subsume

or

knock out each other there's a pluralism
there's

multiple explanations that exist that
capture different parts of how the
system really works

the vindication implies that grand
unified incl uh claims
of the advocates of the ptb are
unwarranted

um

look at what that is again but more
generally we suggest that the form of
scientific progress in the cognitive
sciences

is unlikely to be a single overarching
grand unified theory
little possible indirect

reference to those who claim that the
value or the utility of for example the
free energy principle or active
inference is just simply that it
explains everything i don't need to even
say any more or detail where i used it
or show you the code or show you that
it's better than another model the fact

that this model
i can fit everything into
is the utility and it will subsume other
models and other frameworks
now that's a contentious point that has
to be either empirically supported or
not but it just is so interesting how
dopamine is reward molecule or the other
side of that coin with the punishment
or as a precision molecule the empirical
evidence alone
are not enough to disambiguate that and
so you know trust the data let the data
speak but then this is just showing us a
kind of real time
that our models of the world
get projected as we do sense making on
the experimental outcomes
and um it's not always as clear as it
seems
well you know it just reminds me like so
i mean
just from a very literal sense if we
think about dopamine as um
a reward for minimizing prediction error
like how much do we all like to be right
all of us so i mean like just just for
being correct like that in and of itself
is a huge like dopamine reward and i
think about like i'm a super ultra nerd
and so i geek out on things like
difficult math problems or solving a
rubik's cube or or these kinds of things
and i'm just like finally when i get or
even code right like finally when the
code runs
there's like this huge like something
that you've been working on for a really

long time it's like a huge like dopamine
rush for me which is why like i keep
going back to doing these kinds of
things but but ultimately that's a
prediction error reward
so it's interesting to think about
and here's sort of the two ways that
they're phrased and it's
somewhat subtle but i think it gets at
the difference between reinforcement
learning and active inference
is there a drive for being right
and then we prefer to be certain
or is the drive
for certainty and then there's a
preference over being right or being
accurate
so are there situations where it
actually brings us solace
even though the direction of the outcome
isn't what we expected
or isn't what on some sense conversation
is what we prefer
but actually the fact that we're able to
have certainty
outweighs
the fact that the outcome went the way
it went
so and maybe there's other phrasings too
but it's just again which one do we put
in the driver's seat precision or reward
well so what's the difference between
being certain and being correct like
being right and being certain
like okay if i'm really certain but then
that that certainty is unwarranted
then that's when i'm wrong but then i'll
i have this prediction error gap right

so either way so what's the difference
really between being certain and being
correct

well i think one if i could just jump in
there i think one thing is really
interesting is if everything's based on
energy minimization like you've got
you've got a process you're going to
keep doing it and it's optimizing to
make it basically if i use less energy
i'm being more efficient i'm being
faster i've been smoother so it always
seems like yeah go for the reward the
thing is though

once you get into something where it's
more

complex or uncertain

um

you know the idea and they talk about
this now actually there's a whole area
in um

development work and design thinking
called failing forward fail forward so
the idea is fail fast fail quick but
part of this is like

in

you know and i always say within ethical
bounds right because it might be great
for the designer as long as it's not a
community member being sort of
being frustrated by this but but it
makes sense in the sense that and i
don't think reinforcement learning can
really account for this is the benefit
of doing something

because it might be something which
doesn't work

but now you can take from it it's almost

like
you're
giving yourself
a sample of the counter factuals
in a way by something going wrong but
you don't obviously we can then break it
apart and say well there might be
something right still in that
you know and some bits which are more
wrong and we can sort of
do stuff with that information
almost like succeeding forward would be
better than failing forward i mean
failing forward has this psychological
benefit of helping people recognize that
it's okay to not get it right the first
time and just get something on the paper
and we'll iterate from there so the
rhetoric of fail forward
makes sense but
having something that's even positive
like succeed forward or win forward or
contribute forward or learn forward or
develop forward those actually capture
the functional spirit that um
is here and let's think about how that
relates to blue's question so what's the
difference between being certain and
correct
so there could probably be many takes on
this but
i thought of this classic
science textbook
you know when when they give you the
science textbook um
the difference between accuracy and
precision now in many cases now this
also is not coming from the active

inference ontology so they may be using
these terms in a slightly different way
but the point is we're getting at two
complementary components of precision
and um

basically

uh we can see that there's these two off
diagonals the top left and the bottom
right so high accuracy high precision
it's like your average is on target
so that's the accuracy

is where you're aiming at accuracy aim
and then the precision is referring to
the tightness of the distribution around
that mean

and so you could have high accuracy like
it's always at the bullseye but the
precision is low or the uncertainty the
spread is high

you could also have like a thermometer
with it giving you very tight readings
high precision

but

it's

systematically biased in a certain
direction

and

[Music]

this is just one way to get at two
different aspects now we've seen this in
a few different cases in variational
inference as being the um difference
between the expectations
the means of the distributions
that's like the accuracy
and then the precision is like the
variance of that distribution
so you could have a very tight

distribution

um the the q distribution very tight and close to the p distribution that would be accurate and precise you could also have a really over fit over confident model that is just simply quite divergent from the underlying p distribution

so um which one is being optimized are we trying to aim for a tight scatter and then we and then second order is we want it to be in the middle or do we aim for the middle and then the second order is we want them to be accurate

stephen

yeah that that

to that question is a good one um

and i think distinguish because often they might get conflated in many ways but it's good to pull apart precision and accuracy because it is a

you know you could have

um you know you you could be very precise in the fuzzy set that you have and you could but it could still be fuzzy you could have something like you say more accurate but it moves around a lot and i think that

that's that's an important distinction that might be lost in many cases

yes and it's actually one reason why

it's important to sort of

stick with the uncertainty and pass the estimate

the state estimate as well as like the uncertainty around a state estimate pass that through

because if you collapse the uncertainty

and you treat all estimates as being
equally precise there's this whole
dimension of the data that has been
dropped
for example in gene expression analysis
sometimes there's uncertainty over
whether a gene is differentially
expressed or not like it has a
distribution and you pick some cut off
but then sometimes people take the genes
on one side of that line
and then they just move forward with
them like these are the differentially
expressed genes as if it was a discrete
categorization now take a continuum
and then discretize it which makes
certain kinds of analyses really simple
or straightforward but
a whole aspect of the variability of the
data has been completely crunched out
stephen
yeah and this also then ties into scale
as well um you know you could
and and how much so like you mentioned
the thermometer so the thermometer gives
a fairly
um well fairly precise
sense of the
of of the reading it's fairly consistent
and um it's averaging out the average is
is fairly
relatively accurate in terms of the
average but it's not going to give you
the range or the
the spread within that and that that can
apply to a number of things and then you
as you go down to less and less if you
for instance i suppose like you would

say if someone's purely looking at one
ant
or one ants
behavior at a time
then
you you might have this sense of i'm
being very accurate in terms of my ant
analysis
so to speak however
at the level of the
you know the larger colony
you're you're you're not as accurate you
know so there's some interesting
sort of
scale things that can't be avoided
really
yeah are ants accurate or and or are
ants precise the nest mates
so
it's almost like maybe we could think of
them as being um
high accuracy but low precision like
let's just imagine what that would be
like for the colony so
each forager
is
in the neighborhood of making adequate
foraging decisions but there's a lot of
noise around that so over thousands of
interactions and foraging trips
you're going to end up getting
a lot of your points
in the middle so it's cheap because it's
lower precision so you can have simpler
subunits
but it's centered around the target
whereas if you have it centered over
something that's not the target

it's going to be sub optimally adaptive
for the colony
but i think ants just show that um it's
not about making every single action
perfectly
um
the right one it's about being right
more often than not and then just doing
it many many many times which is related
to that sort of um
develop slash fails succeed forward
sorry go ahead steven
just just one thing i think this is also
useful just in the sense of
and it's almost implicit with the
agent-based modeling approaches that get
used in biology but it's quite
interesting in the sense that that
brings in that inter-subjective piece
what are these ants trying to do for
instance how are they trying to organize
what's going on
rather than the first person perspective
i what do i see
what's my perspective on the ant as an
observer which might be more
psychological in our
world or
what is it that i see the group doing on
average like the temperature
but when you start to get into
agent-based
approaches um
it it does allow this second person um
stuff to happen which is really
i know in psychology is the missing
piece really in a lot of cases
so um you know how this how this in a

way also ties into
thinking about agents
rather than thinking about what i see
someone or something doing from the
outside and
inferring from their behaviors
externally
qualitatively sets up a different type
of um
experiment or modelling
so i wonder about the different
um scales like as we scale from the nest
me to the colony like are
are they comparable like are they asking
the same question like the colony wants
to know like will there be enough food
and the forager never asked that
question the forager always wants to
know where is the food right like so so
they're asking very different kinds of
because the forager knows it has no
uncertainty that that its job is to find
food right and and so the colony wants
to have enough food but the forager
always wants to get the food and i
wonder how this kind of asking different
questions
scales to things like the brain and
components of the brain like
or the internet and components of the
internet like like what are our is is
what defines a scale whether we're
asking the the same question or not like
or or is it the goal of the system that
defines the scale of the system i don't
know it just made me think about
yeah i mean that's a good
that's a really good point i mean yeah

you could if you're starting to look
from the agent
and you could even think about that in
the brain like yeah if different parts
of the brain
are
foraging information
so it's not like that that would explain
a little bit why you have these kind of
parts
you know
you know parts the you know different
parts of the brain have different
personality
um queries going on even when certain
parts of the brain get sort of put to
sleep so to speak so
um
rather than yeah i mean that could be
that could
explain and if they're if they're each
working away i mean
like a colony
um and
yeah who which one is in some of them
get to be in the driving seat
and some just
yeah some get to ask different questions
or some get to sort of have their
questions to influence other questions
and some of them just get on and do
their thing if that makes sense you're a
forager basically you're gonna forage
right that's the deal but someone else
it might be like i'm hungry
you got a forage more
right
oh just i i

rapidly assembling some image memes
it actually breaks down well into the
the chess and other
fields other serious games
of tactics and strategy
so
um strategy so like blue your question
the colony was asking do
we slash i we don't know what pronouns
the colony uses do we slash i have
enough food it's as if it's asking that
question
does it identify as a entity
an i
or does it identify as an ensemble
we don't know but then the nest may is
asking the more tactical
and action-oriented question
which at a first pass is like where is
the food but we know that
in active inference what matters isn't
just like simply inference on the
location of the food out there but what
matters is inference on our action
selection for how we're going to get
that food or how we're going to
implement a strategy that
has a good trade-off of us finding it so
like here
chess meme
um
tactics
mean doing what you can with what you
have so like whatever pieces are on the
board
you're gonna have to do some sort of
action selection for your next move
here strategy requires thought tactics

require observation so it's kind of interesting because observation is one of our terms in the ontology like the agent is receiving observation without observations

tactics cannot be implemented and then strategies requiring mental action just like our previous discussions in 28 the thought is actually the planning over

observation strategies

uh nice double blue

and um there's other definitions of strategy and tactics and how they're similar or different so that's just kind of like

cool that um it's a way to think about maybe not just that

are the levels asking different or the same questions but might levels be defined or delineated by the difference in the questions that they ask

with

independent of the spatial size of the subunit

the more tactical is like more agent level and then the more strategic requiring reflection or cumulative culture or mental action

that level is being implemented by higher and higher

nestings wait blew your mute because you went down to just one

can you hear me yeah

perfect

um i i went down to just one so you can

share the names with us because i'm
missing out on the meme action and i
like i can't stand it a little bit
i'm so sorry
um

so so speaking of action and and
different levels so this is something
that i've really like been thinking
about over the course of the last week
uh this is like the next question on on
on this slide 11.

is this this prospect of
intergenerational stigmergy
and i just wonder like you know as we
were both foraging for memes and slide
content like creating the
29.0 like we have this internet right
and so like we just google search for
whatever and informational foraging i
know is not the same as maybe ants
foraging for you know a
food

but um you know we're both foraging we
come across the same kind of content and
ended up like putting it into the 29.0
in different ways in different places
and and i just wonder you know there's
this concept of of intergenerational
stigmargy that's very uh i
and i wonder if this is human-centric or
if there are other examples of this in
the animal kingdom and perhaps among
ants i know daniel we've talked at
length about

um
the different foraging habits for
example of
different um

different foraging habits of different species or the same species of different colonies

and so where do they get this habit like to forage differently like at different times of the day or whatever and and it spans generations and so is there some trace left in the environment like humans we leave traces for the next generation i mean we directly do some cumulative cultural evolution transfer um and there are other species i know that that teach like dolphins i know teach like some foraging habits and so forth so so

what other examples do we have of this kind of intergenerational stigmergy or like you know we leave traces like the time capsule of you know like what to find like every decade we open up the time capsule from you know whatever the previous decade was and find things like eight tracks and

like whatever things that are put in the time capsule that are just like oh look way back in the day um so i don't know what do you think about that um just relative to ants and relative to this intergenerational stigma g and the different foraging habits of colonies from the same species

good

question

and

uh

where did the priors for foraging come from

and then to what extent is there

scaffolding of one nest mate in their
foraging activity by
older nest mates which is actually
within a generation that's like sort of
some skin cells being scaffolded by
previous skin cells it's like if you
just kill all the foragers of a colony
the foragers that step into those rules
will perform
sub optimally that's actually related to
colony collapse in the honeybees
so there's some element of like they
have an inkling
of how to
adapt their foraging
yet um also there's flexibility um i
mean
it just makes me think of how we could
actually
study it like putting nest mates in
different environments or
having uh
one colony and splitting it and
training it a certain way and then
reintegrating them and seeing if
um

[Music]

what mattered more was like the
environment at that generation or
whether it was possible to have
epigenomic or genomic
changes that sort of lock in and and uh
shape foraging
and then on um
12 you know i'm
doing a little live meme modification
like foraging is not an act but a habit
with a probably misattributed aristotle

quote

so it's almost like forging is not an

act

it's a policy

it's a policy over

specific motor actions

uh

but yeah the intergenerational part is

very interesting

and it

i'm just thinking if there's other true

intergenerational ant scaffolding one

example not in foraging is like when a

leaf cutter ant

queen goes on her nuptial flight she

takes a

little sub-sample of the fungal garden

and uses that to start the next

generation

which

actually i think relates to the second

point you had about 30.

so that's super interesting and also um

directly related to

um the microbe transfer right like you

think about um microbial transfer that

happens um

like through the birth canal like

there's a difference between

c-sections delivered babies and babies

that actually pass through the birth

canal

just in terms of how they're inoculated

as well as breastfed babies versus

formula fed babies

there's a huge like inoculation

difference with your like but just

microbial flora

and so that that is the
uh and i wonder what um you know this
speaks to the holobiont concept
like we're not just humans we're humans
inoculated with all kinds of things and
maybe ants aren't just ants they're ants
plus that fungal and i have there's been
studies on that fungus
like on the fungal composition is it
different from colony to colony
um do you know i don't know
i know that uh one
group in my previous lab they tr they
isolated nest mates and then they
treated them with antibiotics and
changed their microbiome and they
and others have also shown that there's
a role for the microbiome in recognition
so um
because it modifies the smell
of the ants that modifies how they
recognize each other
um maybe that's more like a dialect
like that's kind of how they recognize
that they're in the very short time
range like in the same
group or they're they're likely to be
collaborative but i i returned to eight
to think about what you just described
about like the body and the inheritance
of
microbial forms like
is there a way to refer to that human is
the species but then you know if humans
contain and are
multitudes
you know there's the body there's the
other life forms

there's social

uh interactors

humans aren't the species but the host

which ties in very nicely to our next um

live stream which i'm very excited about

uh

which um

with matthew sims he's gonna come and

talk to us about

uh the symbiosis between

squid and vibrio fischeri in the light

organ

which is something near and dear to my

heart i worked in new mexico i don't

know

uh how is this possible in um right

upstairs you know you have those labs

that like you know everybody in the lab

and you guys can come and work and

um like share equipment sometimes and so

i knew this lab and our lab were very

close uh the lab of michelle nishiguchi

and she studied vibrio fisheries and the

light organ so i knew all these people

doing directly related projects

but that's it's a really cool uh concept

that you know

maybe we're all in some form of

symbiosis and i know in mice they have

sterilized mice and the mice grow up

totally neurotic like mice lacking the

microbiome free mice like they're

completely neurotic and have way

abnormal behavior and they've linked a

lot of you know human behavior or human

abnormal or abnormal behavior or

neuroatypical behavior to microbiome

stuff as well so so it's really

interesting to think about maybe we're
all symbians in some
fashion i mean and like maybe we're not
the species but we're just the host
reminds me of um
demetrius bolus
speaking about the through others we
become ourselves and it was situating
active inference in the dialectical
tradition
and um you have a coming together
of different pieces and they don't
reduce their distinction they actually
become especially over evolutionary time
more polarized like the nuclear genome
becomes more nuclear the mitochondrial
genome becomes more reduced in
mitochondrial or the host becomes
on one trajectory and the parasite goes
in the other they don't converge and
equalize each other they end up becoming
increasingly specialized parts
in
a newly emergent type of system
and then these multi-specified
systems can be fragile because maybe now
there's a failure mode of the part
and then that causes the whole system to
fail so like bucky fuller would always
talk about how
um all cases of biological extinction
are due to over-specialization
might be a bit of an overstatement but
that's sort of the idea is like
generalists can deal with changing
niches better
because they're always able to
hedge their bets

amidst uncertainty and average out over
all the different um foraging
investments whereas a specialist does
really well until they don't but in life
it's not just like you can do well most
the time and then you can fail sometimes
living systems need to
be
succeeding at least adequately all the
time
if any part of your body was under
producing atp for like five minutes
it would get tissue damage so it's
better to be adequate all the time
rather than hyper successful for even
almost all the time
and then what kinds of strategies are
resilient amongst that kind of
uncertainty and situation
so about halfway through what you were
saying i wondered when we started having
a political discussion
because it just really you know speaks
to
specialization or maybe
polarization um which we see in society
so like you know we have these different
factions or different like sub
types or sub classes
of of people and biological systems
that don't
become
more like one another they become
more separate from one another and i
think um you know in the resounding echo
chamber that is um
you know
a self similar political

environment or class environment you
know it becomes this this echo chamber
that that um
you know just increases our
uh
certainty right increases the the
validity of our own model
and it makes us question
the model of others so i wonder
uh and this brings us back to uh what
we're going to discuss with matthew sims
next week
um
is that right or is the jet like is the
generative model
becoming more similar for the fisheries
vapor fisheries in the squid or is it
becoming
more different
i i just wonder yep uh um
so through interaction do we become more
similar or different
or what kinds of systems through
interaction become more similar or
different
and then
how can systems be
more anti-fragile given that
relationship amongst the subunits
versus which which failure modes are you
de-risking on and which failure modes
are you increasing the risk on
like
so it's an interesting question i mean
it's not just political it is very
um broad question
blue let's look at the code
and maybe just scan through some

sections of it and then start to see um
in on 11

before we get to the code it's just like
what else could we include in an ant
model like this was um we talked about
the natural history of the ant that we
templated off of the formica species in
the paper

but um we used one pheromone we didn't
have interactions you know we had such
um a sparse
uh

pulling over of features of ants into
our model so that's why it's just like a
introductory framework so what what
other pieces could exist for ants
and we can kind of use the active
inference

um breakdown of what perceptions could
we add in tactile interactions other
olfactory interactions

light

the

intensity the color the polarization of
light

another type of polarization what type
of cognition could we add in so this was
like a one-step decision-making ants it
just makes a decision for the next time
step based upon the local pheromone
density what kind of cognitive processes
actually exist in ants or what kind
might be interesting to see

in different systems like memory

and then what actions exist

here we had a movement action affordance
any of the directions and then we had
sort of a pheromone deposit affordance

that didn't actually have a specific um
control policy it was just a if then if
you already touch the seed then you
deposit pheromone if you're still
looking for the seed then you don't
deposit fairmont so we can think what
are the perceptions cognitions and
actions of the aliens as currently
modeled in the paper and then where can
we go with that
and then
to go from that question about the
simulation
to meet the empirical data the other
side of the table
what kinds of ant data
already exist
there are several very fascinating
databases
that include genomics
gene expression transcriptomic and
epigenome or epigenomics datasets
they're sparser for ants than they are
for humans and mice but
they are out there and there will be
more of them in the future
what kind of behavioral data already
exists ethology
the study of animal behavior
video data
maybe they analyzed it under a certain
framework but now it could be
readdressed
and then like rfid and qr codes that can
do semi-automated high-throughput
tracking so people attach rfid chips to
honeybees and
track their foraging activity

and then we also have information on
their ecology and morphology so features
of which we're not in our simulation so
this is just this sort of prime looking
at the code

by asking like how could the code be
modified

and then how
could the world

already be seen as reaching out to us
with these helpful pieces of data

and then where are we going to be in the
middle like what kind of model is going
to connect the empirical data sets that
do or could exist

to the models

that do or could exist and then what
questions are most important because we
want to be asking relevant questions and
so it's better to be keeping the goals
in mind as we look through everything
okay let me just

re-crop this window while i get the code
up

anything you would just say on the
overview or like

general thoughts on the kind of
multi-agent simulations or foraging
simulations that you've made

uh just i have some questions about this
simulation actually or some some things
that i would think about to include as
parameters like in a dream simulation
where we you know could can simulate
whatever with unlimited computational
power and

relative world data um so i think that
uh

here i'm interested in nestmate interactions so i think that that would be something that would be interesting to model like the different types of ants that exist like the nurse maids and the foragers and the queen and the uh reproducers i can't remember we went through all the different types of nestmate last time but i would just be interested in in kind of simulating how as a colony ants interact and function i know that that's like you know very whatever big and i mean in most simulations that i've always done it's like the same type of um the same type of agent right like so you give the agent different options but it's never like this agent doesn't forage and this agent does i don't know but i mean i know that they do have different people do some modeling with different types of agents but i just wonder what the compute power is and what the validity of that is yes great okay so we're gonna um take a sort of uh a stepwise approach and one thing that i think will be fun is there's people who know way more about python and coding than both of us for sure including alec chance and others and then there might

be others for whom
they've learned about active inference
or they've read a lot about it maybe
even looked at the math
but here is going to be where we see how
it looks in code so there are these
multiple languages and ways of
communicating and learning by doing
active inference like natural language
math as a language
programming as a language art as a
language maybe there's other ways too
but at the very least we know that we
have natural language
mathematical formalisms like the
equations that we're always putting up
and then here's the code
those
threads might be
looking more similar or different
as we start to learn more about like
interactive notebooks
and ways to
integrate our language use
of the natural kind with the programming
kind
so
we're gonna do kind of a a step down
first we're going to look at the config
file so the readme file in this case is
not super informative which is fine
um but the config file is where we're
going to start that's going to tell us
about like what kinds of variables and
parameters matter
then we're gonna look at figure one and
that's just sort of gonna be like okay
this is how you run it and then we're

gonna pull back to the ans dot py
and look at what functions and
subroutines are actually defined inside
the code
so inside of the config file it's only
30 lines long 31 lines long
and it has
some pretty
clearly defined variables that are being
set
and the idea is that all the parameters
are stored in this file so that you can
kind of hit play and then it will look
to this config file we'll see where that
happens in a second
and so
the first block relates to um
if you want to add the ants in a
staggered fashion how many how many time
steps should happen between adding new
ants in
um
uh how many or what's the location of
where the ants should begin
nest factor we can
look at i'm not exactly sure
um
it is here
it relate i'm not exactly sure where
what the nest factor has to do it might
have to do with the distance of the nest
but
let's ask alec on that one the grid is
the side size of the map so here's a 40
by 40.
um
and then here
is a very interesting block it's the

focal area which is three by three so
that's the grid around each nest mate is
just three cells by three cells
and so this could be changed to you know
like four by four or
something like what happens when the ant
has a broader range of perception than
it might have for action
and that's the action map
and so those are the um the actions that
can be undertaken
relating to basically moving like down
into the left or up into the right
the food location
and the size of the food are provided
then the wall
so that specifies like within the grid
what is the shape of the maze inside of
the map
24 and 25 describe the
the number of discretized levels of
pheromone that exist and then a factor
that is the stochastic decay of the
pheromone trail
and then
here the observation states so like the
o in the active inference model
are
defined as the number of pheromone
levels so in this case
those are both 1 or 10 discrete models
but
it's so interesting because it opens up
the door for the generative process to
differ from the generative model so like
the observations
could be there could be um
three categories of observation low

medium and high
and so you could still have a continuous
variable or discretized variable in the
world in the generative process but the
generative model could be over a reduced
version of that which might be all you
need to know maybe there's enough
stochasticity such that just knowing if
you're in high medium low
tells you whether to put on a jacket or
not you know you don't need five decimal
points on the temperature
and um the number of action states is
also reduced to the focal size but just
as mentioned that could also diverge
and then the um the length of the
simulation i think is specified in 31.
so those are some of the parameters that
exist just in the
base simulation some of the degrees of
freedom that do exist and that's even
before introducing these like other
sorts of um pieces that could exist okay
any thoughts here before we go to figure
one
just like you know i mean i i always
want to know like why
like why did you choose this or like did
you play with the decay factor or did
you play with like the length the time
steps i'm sure like maybe your reviewers
commented that as well but um i just
wanted like i'm always curious to see
like was there any
like
unexpected outcomes or optimal outcomes
that occurred like peak like did you
pick 500 time steps because that was you

know best or the decay factor of 0.01
because that was most real realistic or
those types of things
good questions
and i think we we probably could have um
organized these with some sub headers
and some comments a little bit more
clearly
so
we're going to look at how the
simulation gets specified and then pull
back to like what are the subroutines
so the the top part of a python program
is basically pulling in and then giving
nicknames to other resources that are
required so these resources can include
packages like numpy which is a common
toolkit used for
um
math and statistics in python i'm sure
blue can say more on that one and then
there's this um
the config
as the cf
so that's when we call like cf dot
number of observations
that's going to call like this
which is defined as this which is 10
but of course that can be changed so
that's where the variables that are set
in the config file come into play
so the header
sort of introduces the um
the background libraries and resources
that need to be imported
then
um a few parameters are defined so that
you can have like a base config set

and then you can also just um locally
define some variables like the number of
initial ants is um
like
you can define variables that are in
the config file or you can add some new
variables in
and then um
basically
uh
the
uh
for
the observations
in the length of the range of
observations
this runs
a routine
and um
the routine is going to be defined
better in and stop py
but it basically says we're going to run
it
for that many steps for that many
starting ants for that many maximum ants
we're gonna save it
we're gonna switch the location of the
food
and then we're gonna just print some
summary descriptors
like the number of round trips
uh
and then
save that
in
[Music]
a file
so that's like sort of the

interface wrapper
for running a simulation and then
outputting some of the descriptive
statistics and then also saving
that file
now ants.py is
353 lines long so
um it's a little bit longer
but there are some parts that we could i
think walk through
the first is that like
the ant is a class
of object
and it has um a few routines
it has its initiation routine
which is where its x and y position for
example are given
the starting location
and then there's updating a forward and
a backwards step
there's the class of the environment so
this is kind of like if you do net logo
modeling agent-based modeling you know
you have like the turtle like the agent
and then the environment so here we're
defining the agent and we're defining
the environment
and the relationship between the two of
them is really whether we have um
a stigmatic type multi-agent system or
not
so you could have the environment
influencing the behavior of agents
like the sunshine is influencing the
behavior of the ants but then the ants
don't modify the sunshine so that
wouldn't be stigmargy but
if there's a possibility for the agent

to modify the environment and for the environment to modify the probability of agent behaviors there's some sort of bi-directional relationship then it is stigmargy

and um

basically some of the functions that are defined for the environment

have to do with uh

which

observations are at a given

cell

whether the food exists at a given cell

whether the walls

exist at a given cell

and then the step forward a step

backward relate to the outgoing and the

homecoming

trip and this is basically just um

[Music]

uh

doing a delta which i think is a which

is the action map it's the delta in

position

and then it checks whether the ant is

returning which is like a gonna be i

believe on the step backwards side

and um

[Music]

versus the ant

going forward

decay is a function that's also inside

of environments

because decay has to do with the decay

of the pheromone at that cell

and then the plotting has to do with

just saving

um the file so that's some of this like

uh the gif and the save fig
another object type that's defined and
this is really where there was the main
carry over from the previous models is
the markov decision process
so here just like we've talked about in
various contexts like the a b c and d
are all
uh
defined and initiated here
those are um
used as you'd expect
and then the mdp needs to have these
functions that relate to setting and
resetting certain variables
and taking a step and then i love this
line 210
action inference like this is sort of
the kernel
that
is where it's happening
here's the negative expected free energy
and then there's a calculation
of the negative expected energy of the
uh negative
utility calculations
divided by the scale factor
pretty interesting and i don't think we
experimented with modifying this scale
factor
and then um so can i just interrupt you
what does that scale factor do really
well
it's mentioned only two times in here
and
um
i'm not exactly sure what it does here
i think that uh when you're comparing

relative policies

as long as you're dividing them all by
the same scale factor

it shouldn't change the relative
preference for different policies if
they're all being scaled down by the
same level

so

i'm not sure why the calculation exists
this way

yeah that i mean it makes sense that
they're all like whatever you scaled by
something maybe it's just to make the
numbers more manageable who knows
yes

but we'll let's definitely think about
that

and then um

[Music]

it yeah

we could get more explanation from alec
um but

the uh

negative expected free energies
get so

okay well okay one possible
option would be this is like over
observations

so we know that the observations are
discretized between one and ten
so if we divide

by 10 maybe we're going from like a 1 to
10 scale to a 0 to 1 scale

which then lets us interpret
more cleanly as like probabilities
because

um the action

ends up being selected based upon the

soft max of the negative expected free energies

so instead of like you know eight one and one

if you do

0.8 0.1 and 0.1 it's really easy to then say okay right 80 likelihood of choosing this action

here's a colony level function related to an ant being initiated and just appearing on the map plotting and saving functions and then here's the main

so um

a lot of times in the python programs

there's like a lot of sort of upstream functions that are going to get called and then main is called and then when main is called that's like where the main program actually happens

so this instantiates the environment the ants and the paths

it for the number of ants in that are to be specified that many ants are created with reset

mdps

all of the image files are set to zero the descriptive statistics like how many completed trips and the ant locations are reset

and then here

is the time point time step based simulation so here

for each ant in the ants that exist there's going to be some calculations this is related to the distance calculation

um

here's adding the ants

here's switching the food

and then here is um checking for each
ant trajectory

whether it's returning or completed
and then saving the file

so

again i think there could be a lot more
organization and commenting
to sort of understand what these
different pieces do but
yeah i believe that's some of the main
pieces

nice

are we gonna move to the figures now
um

sure i can actually try to download the
videos and i can play the videos

oh yeah that's great so yeah i thought
we were going to go switch back and
forth from the code to the

okay so we'll go to the paper
supplemental data

let's see if we can play it in the
stream directly

will it play

oh great it will play okay so here is
like the ants

you'll you won't see it blue but um
diffusing around the red ones are going
out and the blue ones are coming home so
this is just as if you were um watching
the colony with video

um video two

i think we're gonna add in
the pheromone density so here the
brighter colors are the pheromone

density

and like there like one ant made at home

but then it didn't

here comes another one

it didn't um

[Music]

sort of lock on but now we can see

there's a light ridge

it's locked on

to the trail and it's going to reinforce

the trail

and then

it seems like the food has switched

because now we're seeing some pheromone

getting deposited on the left side

and then there's not as much deposition

on the right side but this is what

really happens in ants like if there's

high density of pheromone on that top

right

then

an ants are spending a lot of their time

there

especially if they're carrying food they

might actually be reinforcing

that part so some of the videos are

described their similarities and

differences

in the text but i think the interesting

one to look at would be

i think six and seven

so here

there's um here there's a preference for

high pheromone density

and then

i believe it's seven

where the ants can lay down pheromone

but they don't have a preference for it

and so it ends up getting laid down but
then it doesn't get reinforced because
they um
they just don't prefer it so they act
randomly with respect to pheromone
um
but these are just like fun
simulations and yeah a total
obvious next step an important one would
be to
explore what sets of parameters certain
outcomes occur within like when do they
always lock in when do they fail to lock
in
um
but yeah good
supplemental info
awesome
um
yeah so what what is the next obvious um
step for you or where would you like to
see this simulation go
most like what is the the next if you
were to you know
have unlimited time with alec to say
like let's code xyz
um what what would you uh dream of
first i'd probably go for the annotation
and clarification
because that's like where coding
expertise really comes into play and
that helps other people
modify and build on the code the most
um
when it were a little more reorganized i
think it would make sense where other
sensory and action affordances could be
plugged in

but the piece that's
such as like adding another pheromone so
instead of just having like you know
pheromone one through ten density like
make it specific pheromone sent one
and then that would allow us to go okay
now let's just copy and paste pheromone
sent two
and then we could have
um more flavors
so well so what what would you make the
pheromone scent to
do would you make it like fairmont said
one drives you to the food and if
everyone send two drives you home or
would you just make different decay
rates or what would you do with the
different fair ones
one duality would be like plus and minus
if there was the appropriate behavior
around using them
another one would be fast and slow decay
which maps on to the way that ants
actually use the sense with like some
longer chain
hydrocarbons that are longer lasting and
some shorter chain ones that are
volatile
um another interesting cognitive model
would be like there's two or three
pheromones
and then what the inference is on
is the ratio
and then it could be shown like you know
no cheating the ant is only perceiving
the ratio of these two things
so that would be kind of
uh

it's kind of like bimetalism
like you tie to the idea of a ratio
rather than to just the absolute amount
of one
so like
um

it could be food could be a little sweet
and a little salty or a lot sweet and a
lot salty or then the inference of
whether it's tasty is like whether it's
balanced

so then if we're thinking about the ant
tasting

balances in the world

maybe that could help it have an
internal manifold

that by definition would be simpler
than the state space of the pheromone
densities out there in the generative
process

um

so i i heard that there are two types of
ants like that there are grease eating
ants and sweet eating ants are they just
like different

like what is that what makes that
distinction actually and like does the
same ant really eat those things or
what's up with that

um

just reminded me of you know like uh uh
evil neutral good like the
classifications for characters
and then what are the major macro
molecules what are the nutritious
molecules other than vitamins well
there's like protein carbohydrates and
fats lipids

and
different parts of the colony life cycle
might use these different nutrients
differently but then what are the
resources that could be obtained like
sweet that could come from like an extra
floral nectary so like a plant that's
secreting a honeydew or scale insects
and the the carbohydrate-rich fluids
that they secrete and ants that love
sugar and pies and stuff
then there's oil
and then there's also like protein like
there's meat ants

and um

the turtle ants

in the in the canopy forests in some
rainforests i've heard from many ant
researchers like that the most effective
traps

use human urine

and like those ants well that's high
nitrogen high protein relatively
speaking if you're a tiny guy

so

um

[Music]

it's interesting why some species would

um

prefer

more or the other if you're putting on
more biomass

like if you're making larger colonies
you probably need a lot more protein and
lipids

whereas if you have longer lived workers
then you don't need as much lipid and
protein but you might use a lot of

carbohydrates

so that's sort of the fun way to think

about the workers like a car like you

need the metal to make the car

but then you only need gas it's like you

need the protein and lipids to make

the ants

but then afterwards you really use more

carbohydrates

even though they probably also digest

other nutrients but i think the the

piece that um to connect would be like

um

find some

student or person who loves

meta-analysis it could be us or it could

be someone else

and establish like which data sets

are already there

what's a recent impactful and

publication that already has done a lot

of the meta analytic work

of at least compiling data sets and

there are several ants databases that

have compiled behavioral data

morphological data ecological data

genomic data

and then um

how could that

data format play into

this

model

what are we going to

put in

and then i think uh a good step would

just be to show like

what does the octave inference

ant model

explain predict suggest

that

uh

other first principles approaches like

reinforcement learning might not

perceive

very nice

yeah so my question about the different

types of food that ants eat was was

really

linked to different pheromone

um

like types or ratios or concentrations

like if i have a high preference for

lipids but i'll eat sugar like

you know maybe my ratio preference is

different like this is another way to

like enable different like types of ants

or just to give them different

preferences over the food that they eat

and then where like if we initiate the

same kind of agent and the same kind of

food did they all like proportionately

distribute or do some carbohydrate

referring ants go to lipids instead of

lipid preferring antibiotic

carbohydrates i just wonder and like

does it have to do with proximity or the

number of ants that were there before or

i think that all of that is is

interesting and kind of relates to

like this multi-goal

aspect of active inference it's always

really kind of difficult to

i mean it's easy when we think about and

get food right like so the active

inference model is as easy it's

manageable is what it is so it becomes

computationally tractable and like
mathematically tractable when it's a
single type of agent with a single goal
or even like two different types of
agents with two different types of goals
that are overlapping i mean so you can
see just how but it exponentially
expands when you have this kind of
multi-goal
thing and like some goals are like more
pressing than others right like my
immediate needs have to be met for me to
like you know desire to go to college
like i have to have food water shelter
clothing etc
and so these kinds of
multi-goal and like prioritizing goal
setups
are um really interesting and
a great way i think to start to
approach
modeling real
or
us
or
complex
systems that have multiple conflicting
sometimes conflicting goals
another setting to transpose and explore
that where there's a ton of data would
be first make active beefarence with a
honeybees so there they don't use stigma
g
um on the trail because they're on the
wing they're flying
but in the nest you could have like us
in the hive as it were you could have a
special dance floor and so that is where

maybe like a active inference
communication model could come into play
so
that this would be less on the stigmargy
side but more on like the multi-agent
distributed communication side so flip
out the trail um
modification with a symbolic waggle
dance
whether it actually does a waggle dance
or whether we just say okay there was a
waggle dance that occurred and it had
this precision and these estimates
then
honeybees are well studied for making
the trade-offs between foraging for
nectar carbohydrate pollen which has
like lipid and protein and water
and
a lot of work on like
is it that um
a hundred percent of the variance in
that foraging outcome being different is
about the needs of the colony and all
workers respond to the needs of the
colony
is it that
the workers have their own nest mate
level preferences
and then the colony outcome just is what
it is or is it somewhere in between
where there's preferences that differ
between workers maybe as a function of
their genetic backgrounds but then the
colony outcomes are also in feedback
with them
so that could be um a case
and i think this kind of takes us back

to um
to slide eight which is like when we
have
the systems lined up
and we understand their similarities and
differences
then
we'll be able to transpose
across different systems
in a really interesting way and then ask
um it's kind of like biomimicry on
behavioral systems
is this more like a honeybee situation
where people are exchanging information
and then going their own way
or is this more like a trail stigmergy
situation
and um
so like there's going to be some
early work transposing into these major
different domains like again this was
the first paper that transposed active
inference and included stigma g although
that had been discussed by axel constant
and others
regarding ecology it had always been
qualitatively a part of the discussion
that niche modification was relevant and
that agents were able to be in feedback
with their surroundings
and then it went from the natural
language
through the formalism to the code and
then now it can feed back change how we
think and write and speak and then we'll
transpose into a few of these major
domains start to sub-specify into
situations and it'll be very

cool so it's super interesting about
like the um communication versus like
the trail like which of these like
and i just wonder what you think like
as humans
which are we more like are we more like
bees or are we more like ants like do we
communicate and then go on our separate
way or do we just like follow in a row
like
where do you think we're at
in the b ant trajectory
it makes me think of a drive that we
went on this weekend and um there was
the street signs and then there was a
big field and then there was the
pleasure path that was cutting the
hypotenuse across the field so it was
like a big block with an empty field in
it and so people were walking and so
they were making that trail and so there
was this juxtaposition between the um
[Music]
societally imposed and scaffold
infrastructure like the streets and the
symbolic signs you know the stop sign
here and this is where this street and
that street intersect and then there's
the people
who were moving on that landscape and
then even if you don't see any other
people at that moment it's their
stigmargy with the trail
and so it is an interesting question and
it's almost like we have symbolic rules
and scaffolds like law and norms and
culture
that are even though there's ways to

think about those as stigma energy
themselves
for the agent level that's more like a
symbolic communication
and then
it's like the ant stigma g is like
blurring the lines
that the symbolism provides for us
so
i think there's um
there's some components of both plus
other insects
maybe not insects too but that's quite
contentious so that that brings me back
to like the concept of of
intergenerational sigmoidy but but
really like maybe my question was not
like phrased correctly like sure we
leave some stigmatic traces as humans
for like the the next generations but
but also like we leave
like sigmoidy for strangers right like
like so i wonder if like and i know that
the ant colonies like the pheromones are
different and the different species of
ant they're different and so i wonder um
maybe is it like the fact that total
strangers can go you know searching for
um collective intelligence or collective
behavior or something like that on the
internet and come to the same result or
certain like total strangers see the
stop sign and know that there's some
sort of behavioral norm that's supposed
to be enacted just by seeing the signs
in the street and so we leave this like
you know we do this externalization of
of cognition a lot as humans and so

this is what i want to see in the animal kingdom like like i want to see traces of like can strange animals say like oh i see a rock stack i know the trail goes this way right like like does that happen in the animal kingdom or is that what's maybe uniquely human good question and i think this is why we can study these systems and gain insight without naively copying their strategies or thinking that we must be them or they must be us because our whatever it is that we're doing it's at the intersection of like stigma g and cumulative culture modification and symbolism so in a way the ant and the bee foraging paradigms not no hashtag not all ants because some don't do pheromone trails etc but one extreme is like the syntax of the foraging trail and there's no symbolic communication it's just stigma g and then the other extreme is no stigmatic modification there's no trace left in the environment at all there's just transient symbolic communication so it's this continuum from the transient symbolic like i'm going to pass you a message and then you're going to be off on your way to the stigmatic and often sub symbolic like the path crossing that field it isn't directly a symbol

saying walk on me
but
it isn't
in affordance we perceive it as an
ecological affordance but it doesn't
have it that's not what it symbolizes
that's what it is and that's how we
enact it um so like we could have some
types of active inference models that
highlight
that um
transient among agents
meaning based communication
versus models that um
don't need to take on any grandiose
hermeneutic
communication and what does
communication mean and how does thinking
through other minds happen
we just
look to the ground and about how agents
respond to their niche which is a
combination of biotic and abiotic
nice
fun times
yeah just any final thoughts or or any
anything else for me i've given all my
all my thoughts out i think oh me too
over the years but thanks for um pushing
for this paper for 29
and uh 30 is gonna be
hot on the heels slash
abdomen of this paper because i think
we're going to be able to operationalize
a lot of the discussions that we had
about extended cognition i know it's
also going to bring back
what we talked about with the

definitions of a self and individuality

so

another one in the books on to our third

decade

of um

papers on the live stream

and uh

yeah i hope people found it interesting

as with other

papers except this is one where

hopefully they can

get in touch with the authors and

understand that it's a totally open area

and a collaborative one too

awesome well thanks for giving us your

time and and always giving us your time

thanks youtube luke so

um i'll flip to the closing slide

that's

what we can still think about

and uh

awesome see everyone around next time